

Technical Supplement: Activation-Aligned Asynchronous Residual MPC

for Position-Controlled Manipulators with Learned Dynamics

Public evidence bundle for ROBIO 2026

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1 Purpose and evidence boundary

This supplement provides implementation, configuration, and diagnostic detail compressed in the conference manuscript. It introduces no results beyond the frozen or deterministically reanalysed public evidence bundle.

The simulation results use MuJoCo position-controlled robot models. ABB IRB 2400 is the primary robot; UR5e is an independently trained and calibrated nominal within-robot replication. The bundle also contains a separate SO101 physical case study with a hardware-specific model and controller calibration. The evidence does not establish zero-shot sim-to-real transfer, hardware-independent generalization, external Cartesian accuracy, ABB hardware-torque feasibility, hard-real-time scheduling, robust-MPC guarantees, shared-model transfer, or arbitrary-morphology generalization. The post-freeze effort sweep is diagnostic only.

2 Controller and execution configuration

The controller is a one-layer GRU (hidden size 256; history length 16) that predicts velocity increments from aligned $[q_t, \dot{q}_t, q_t^{\text{ref}}]$ tokens, using one-step increment plus 20-step rollout loss on a 90/10 episode-disjoint train/validation split; recurrent windows remain within their assigned episode groups. Frozen controller cost weights/scales are shared; XML models, data distributions, checkpoints, normalizers, references, Robot-Spec contracts, and delays are robot-specific. The datasets are independently collected offline closed-loop position-actuator transitions driven by non-MPC reference excitations: hold, step, smooth-random, sinusoidal, delta-reference-random, and temporally correlated random references. The latter uses an autoregressive random-reference generator; no training rollout is generated by CEM or MPC. Workspace bounds, normalized excitation scales, and optimizer settings remain robot-specific. For UR5e, the frozen dataset contains 120,000 episodes of 1,000 control steps (120 million transitions) collected at normalized excitation scales 0.3, 0.5, 0.8, and 0.9. The complete machine-readable contracts are released with the evidence bundle.

Table 1: Frozen ABB controller settings. Per-joint vectors and all robot-specific artifacts are in the manifests.

Setting	Value	Setting	Value
Control interval	10 ms	History length	16
Horizon / samples / iterations	20 / 128 / 2	Elite ratio	0.08
Replan interval / delay D	5 / 6 steps	Guard	5 ms
Temporal discount	0.95	Projection	two-stage compiled
$w_q/w_{\dot{q}}/w_r/w_s$	1 / .10 / .20 / .05	$w_{\ddot{r}}/w_{\dot{r}}/w_{\text{first}}$	.05 / .02 / .20
Joint / velocity barrier	10 / 5	Barrier max multiplier	2
Task position/orientation	1 / .25	Task scales	.05 m / .0873 rad
Feedback $K_q/K_{\dot{q}}/\delta r_{\text{max}}$	.3 / .015 / .015 rad	Residual maximum	(0.12,.10,.12,.15,.15,.20) rad

3 Physical SO101 case study

The physical result is a deployment validation of the residual-planning principle, not a direct reuse of the six-DOF simulation hyperparameters. The SO101 follower controls five arm joints at 30 Hz. Its separately collected GRU uses 16 history tokens $[q, \dot{q}, q^{\text{ref}} - q]$ and predicts a direct Δ -state target. The frozen hardware

instantiation uses $H = 6$, 128 CEM samples, two iterations, residual authority $|r_j| \leq 1^\circ$, and the tracking-dominant objective $J = C_q$. The executable-command projector, joint/velocity/acceleration/braking constraints, encoder quantization, and startup/homing gates remain active. The formal matrix contains 54 held-out trials and nine development-circle trials, with Direct IK, fixed Preview6, and NN-MPC in rotated pre-registered order. Direct IK denotes direct execution of the frozen IK-derived joint reference; Preview6 advances that same nominal by six control steps. These predictor parameterizations are hardware-specific, while residual CEM and asynchronous packet semantics are unchanged. All technically complete trials are retained.

The primary endpoint is encoder-space joint RMSE over the frozen `SEGMENT_SHAPE_LOOP` window. TCP values are forward-kinematics calculations from encoder configurations using the calibrated fine MuJoCo model; no external Cartesian tracker was used. The public aggregate and sanitized trial ledger are available under `so101/analysis`.

Table 2: Held-out SO101 physical aggregate. TCP is FK-derived and secondary; command activity uses transmitted executable commands.

Method	Joint RMSE (deg)	FK TCP RMSE (mm)	Cmd. vel. RMS (rad/s)	Cmd. acc. RMS (rad/s ²)
Direct IK	0.910	8.912	0.0435	0.7699
Fixed Preview6	0.675	6.538	0.0435	0.7701
NN-MPC $\pm 1^\circ$	0.353	3.715	0.0807	1.1985

Across the 54 held-out trials, NN-MPC reduced joint RMSE by 61.2% versus Direct IK and 47.7% versus Preview6. It won all 18 matched held-out pairs against both baselines; the paired means and confidence intervals are recorded in `so101/analysis/public.summary.json`. Across 20,956 physical planner events, mean/p95/p99 latency was 22.73/28.36/29.93 ms (maximum 68.09 ms), with three late-dropped solutions and no control deadline miss, planner failure, or runtime safety violation. This timing result establishes deployability and does not independently isolate activation-alignment necessity on hardware. Diagnostic acceleration exceedances induced by encoder-command quantization are reported separately in the trial ledger and are not executable-projector safety violations. Command velocity and acceleration RMS increased by 85.4% and 55.7% relative to Direct IK, respectively, reproducing the simulation tracking-activity trade-off.

Injected-delay stress. We additionally evaluated the same `ThreadedAsync` execution path with a deterministic 33-ms delay inserted after packet post-processing and immediately before publication. Controller, model, reference, CEM, residual, projection, and safety settings remained frozen. An independent shadow pass produced 2,327 finite latency samples and selected $D = 4$ from the measured p99.5 latency (100.601 ms) plus a 5-ms guard at 30 Hz; the resulting late-drop and packet-expiry rates were both zero. The v2 OOD envelope covered 100.000% of executed histories and 99.683% of both selected actions and predicted states. The preceding v1 calibration was excluded because it had 1,734 samples and a 2.768% late-drop rate.

Table 3: Paired SO101 injected-delay stress results. Each row is a held-out fast-ellipse phase block; the final row is the arithmetic mean across blocks.

Block	Baseline joint RMSE (deg)	Delay33 joint RMSE (deg)	Difference Delay33-baseline (deg)
ellipse_fast_p0	0.34685	0.41896	+0.07211
ellipse_fast_p1	0.31772	0.45074	+0.13302
ellipse_fast_p2	0.33127	0.42876	+0.09749
Mean	0.33195	0.43282	+0.10087

Across the three matched blocks, planner-event timing statistics were computed after concatenating all events within each condition. Baseline pooled latency was 22.56 ms (mean), 28.33 ms (p95), and 29.37 ms (p99); under delay it was 79.32, 94.34, and 99.44 ms, respectively. The stressed p95 spanned 2.83 control periods. All six runs completed without control-loop deadline misses, planner failures, late drops, packet expirations, command-limit violations, or runtime safety violations. The delay condition was worse in all three pairs; the paired Student- t 95% interval for the mean increase was $[+0.02488, +0.17687]^\circ$ (three pairs, $df=2$). Encoder-quantization acceleration diagnostics are reported separately and are not executable-projector safety violations. This stress test supports physical deployability and latency sensitivity of the aligned execution path, but it is not a hardware `NaiveDelayed` or no-alignment ablation.

True-history window validation. Held-out model errors use fixed windows beginning at steps 16, 36, $\dots$, 176 of each saved 200-step rollout. Every window therefore has a complete 16-token ground-truth recurrent context before its first predicted successor. Each horizon is evaluated as a separate segmentwise RMSE: $h = 1$ measures the first successor and $h = 20$ pools prediction steps 1:20 within each window. These are not

terminal errors. The historical start-of-rollout validation, which padded the missing initial GRU history, is retained in the evidence bundle for audit but is not used in the manuscript table.

Residual packets encode corrections, not reusable absolute commands. At packet age $\kappa = t - t_a$, the full protocol reconstructs and projects

$$q_t^{\text{cmd}} = \Pi_{v,\text{acc},\mathcal{Q}}(\bar{q}_t^{\text{IK}} + r_\kappa + \delta r_t; q_{t-1}^{\text{cmd}}, \dot{q}_{t-1}^{\text{cmd}}).$$

The launch-anchor replay counterfactual instead projects the cached planned absolute command q_κ^{plan} . Its large failure is therefore evidence for execution-time nominal-frame reanchoring required by an independently evolving nominal, not a competitive controller comparison.

Optional post-selection model-disagreement gate. The released controller exposes an optional gate that is disabled (`uncertainty_mode=off`) in every reported configuration. The planning GRU alone generates and scores all CEM candidates. Only after selection, independently trained, architecture-matched GRU dynamics models roll out the same selected H_u -step joint-reference sequence; all use the same non-MPC reference-excitation training protocol. If $\hat{x}_{h,d}^{(j)}$ is the prediction of model j at step h and state coordinate d , and s_d is the planning model’s training-state standard deviation, the gate records

$$u = \sqrt{\frac{1}{H_u n_x} \sum_{h=1}^{H_u} \sum_{d=1}^{n_x} \left(\frac{\text{std}_j \hat{x}_{h,d}^{(j)}}{s_d} \right)^2}.$$

This post-selection calculation neither rescales CEM candidates nor changes their costs, ranks, or packet activation step. In monitoring mode it is telemetry only. When the optional gate is enabled, a hysteretic rule can attenuate residual authority after sustained disagreement. An incomplete time-bounded evaluation, or high disagreement together with a predicted joint-margin, residual-saturation, or tracking-error-growth condition, invokes a strict fallback: both residual and packet feedback are set to zero, yielding the same Projected Direct IK command. A limited drift state can instead use current-reference feedback rather than packet-predicted feedback. Thresholds, model count, short horizon, and evaluation deadline are configurable implementation parameters; this paper does not calibrate or evaluate them and claims no OOD-detection capability, closed-loop benefit, or safety guarantee.

4 Artifacts, references, and calibration

The manifests record robot contracts, command limits, model/data identities, references, and activation delays. The public calibration payloads in `abb/calibration` and `ur5e/calibration` use

$$D = \left\lceil \frac{L_{\text{P95}} + g}{\Delta t} \right\rceil,$$

where L_{P95} is snapshot-to-publication latency, $g = 5$ ms, and $\Delta t = 10$ ms. Independent 500-plan calibrations gave ABB $L_{\text{P95}} = 52.28$ ms and $D = 6$ control steps, and UR5e $L_{\text{P95}} = 56.94$ ms and $D = 7$ control steps. For UR5e, `ur5e/contracts` additionally exposes the robot and reference contracts.

5 Full diagnostics

Table 4: Held-out open-loop model errors from complete-history windows, pooled over prediction steps 1: h . σ_u is normalized reference-excitation standard deviation. Errors are $q/\dot{q}$ RMSE in rad/rad s⁻¹; Div. is the divergent-window rate at $\sigma_u = 0.8$.

Robot, h	q RMSE, $\sigma_u = .5$	$\dot{q}$ RMSE, $\sigma_u = .5$	q RMSE, $\sigma_u = .8$	$\dot{q}$ RMSE, $\sigma_u = .8$	Div., $\sigma_u = .8$
ABB, 1	.000670	.001990	.001069	.005570	0%
ABB, 20	.004852	.034472	.006876	.051398	0%
UR5e, 1	.000132	.000465	.000160	.000532	0%
UR5e, 20	.001334	.002745	.001436	.003023	0%

Model validation and CEM ranking. Tables 2–3 provide the principal held-out and replay diagnostics. Complete window, rollout, excitation, and counterfactual-replay files are released in the evidence bundle. Replay covers retained, projection-consistent online candidates only; its lower branch error is not directly comparable with the broader random-excitation validation.

Table 5: ABB retained-candidate replay diagnostics. Spearman is computed within snapshots with defined ranks; all intervals are hierarchical bootstrap 95% intervals.

Metric	Estimate	95% interval
Within-snapshot Spearman	0.976	[0.917, 1.000]
Pairwise concordance	0.988	[0.957, 1.000]
Relative realized regret	0.192%	[0, 0.481]%
20-step branch q RMSE	0.889 mrad	[0.705, 1.121] mrad

Supporting planner diagnostics. Task-space reranking reduced deployed ABB TCP RMSE by 3.05 mm (95% CI [0.96, 4.32] mm) relative to joint-only final-pool scoring, while increasing end-to-end p95 latency by 6.82 ms. Two-stage compiled projection met its pre-specified 5% non-inferiority criterion against full compiled projection. These are accuracy–latency diagnostics for supporting execution mechanisms, not independent core tracking claims.

Robustness and timing. ABB robustness applies L3/L6 single-factor payload, actuator-gain, TCP-force, and observation-noise perturbations. Per-condition/bootstrap results, timing, packet behavior, projection activity, and paired Projected/Preview-IK comparisons are released with the evidence bundle. Primary experiments had no planner failure or imposed joint, command-velocity, or command-acceleration violation; this does not imply a physical actuator limit.

Mechanism isolation. The frozen 20-case ABB matrix gives +2.82 mm for removing activation alignment (95% CI [1.27, 4.39] mm), +654.60 mm for launch-anchor absolute replay ([528.68, 778.68] mm), and −0.95 mm for removing fast feedback ([−2.25, 0.28] mm), all relative to FullVirtual. A focused recurrent-history isolation changed TCP RMSE by −0.67 mm with an interval crossing zero. Thus the evidence supports activation consistency but does not establish an independent nominal tracking gain from history propagation. Supporting projection/task-score and post-freeze effort diagnostics did not replace the frozen primary configuration.